

AgriSense: An Integrated Multimodal Framework for Crop Yield Prediction and Pest Risk Intelligence in Precision Agriculture

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Abstract—Crop yield and crop health are jointly determined by weather, soil, environmental, and geographic factors acting together. Most existing prediction tools, however, examine only a single facet of this picture. A model built purely on weather, or purely on remote sensing, is rarely able to explain why a given field is underperforming or which factor triggered a specific alert, and this narrow perspective constrains how useful such tools can be for whole-farm decision-making. We introduce AgriSense, a multimodal framework that consolidates weather, soil, remote-sensing, terrain, crop, and spatial data into a single location-aware pipeline built around two connected tasks: crop-yield prediction and pest-risk assessment. LightGBM is used to predict yield, while Random Forest handles pest- and disease-risk. Growing Degree Days (GDD) are incorporated as a derived feature, giving the models explicit access to temperature-driven crop development that raw weather variables capture only weakly. A GIS layer links both sets of predictions back to farm locations, allowing a yield estimate or risk label to be read directly off a field map rather than from a table. We further examine several combinations of modalities and models to gauge the contribution of each input. LightGBM provides MAE, RMSE, and R2 values of 0.27, 0.38, and 0.93 for yield prediction respectively, which beats the baseline models Linear Regression, Random Forest, and XGBoost, while Random Forest performs the best among the classifiers analyzed for pest risk prediction, providing accuracy, precision, recall, and F1 scores of 0.91, 0.90, 0.89, and 0.89 respectively. All the aforementioned factors contribute to building an integrative and interpretable basis for agriculture decision-making.

Index Terms—Precision agriculture, crop yield prediction, LightGBM, Random Forest, pest risk, GIS, multimodal learning.

I. INTRODUCTION

Yield is determined by the interplay of various elements such as weather, soil, crop characteristics, environmental factors, and location, and it is the combination of all of these that makes the problem of predicting yields and monitoring crops difficult. This challenge is particularly evident among smallholder farmers and mixed farms, because none of the sources of information provide a complete picture of a field. This is where machine learning can be of great help, as it enables taking into account several factors at a time instead of applying a single heuristic approach. According to Van Klompenburg et al., who surveyed the topic, weather and soil characteristics were the two most frequently used features for modeling, and the performance of a model mainly depended on the quality of the dataset [1].

Several recent studies back this up in different ways. Gupta et al. forecasted rice yield with a neural network built on temperature, rainfall, and humidity data [2]. Pathak et al. went further, combining satellite imagery with weather, soil, and terrain layers, and found that how much each source mattered changed a lot depending on the crop and the region — which led them to argue against reusing one fixed model design everywhere [3]. Work on crop health reveals a related shortcoming. In their review of this class of methods, Domingues et al. found that most deployed systems remain narrow in scope, each restricted to a single crop, pest, or sensor [4]. Babu et al., and G. et al. for maize, take a similar single-crop approach, pairing IoT-based soil monitoring and web-delivered advisories with conventional learning algorithms for disease prediction [5], [6].

Growing Degree Days (GDD) offer an additional useful signal. They track the accumulation of temperature and, with it, crop development in a way raw weather variables do not capture. Keramitsoglou et al. developed a service that estimates field-level GDD from satellite temperature retrievals, matching ground-station values closely enough to support real phenological decisions [7]. Separately, GIS-based platforms have been proposed to consolidate heterogeneous farm data and support management-level decisions in precision agriculture [8], [9].

Taken together, this body of work points to three recurring gaps. First, most systems couple a single data modality to a single prediction target, making it difficult to explain why a yield or risk estimate shifts when conditions change. Second, pest-risk and yield-prediction pipelines are typically developed and evaluated independently, even though pest pressure and yield outcomes are agronomically related. Third, few systems carry a prediction through to spatial decision support, converting a number or risk label into something a grower can act on for a specific field. AgriSense aims to address these three gaps jointly rather than in isolation.

The framework merges weather, soil, remote-sensing, terrain, crop, and spatial information into a single feature space. LightGBM performs crop-yield prediction, Random Forest performs pest and disease-risk assessment, and GDD is added as a derived feature. A GIS layer connects both sets of predictions to farm locations, so that a yield estimate or risk label appears on a field map instead of remaining an abstract table entry. We also

assess several combinations of input modalities and models to determine how each contributes to prediction performance.

The remainder of the paper is organized as follows. Section II describes the data, preprocessing pipeline, feature engineering, and the two prediction sub-systems within AgriSense. Section III presents the crop-yield and pest-risk results along with the multimodal ablation strategy. Section IV addresses implications, limitations, and threats to validity, and Section V concludes with directions for future work.

II. MATERIAL & METHODS

A. System Overview

Fig. 1 shows the AgriSense pipeline from end to end. Six data sources — weather, soil, remote sensing, terrain, crop, and spatial

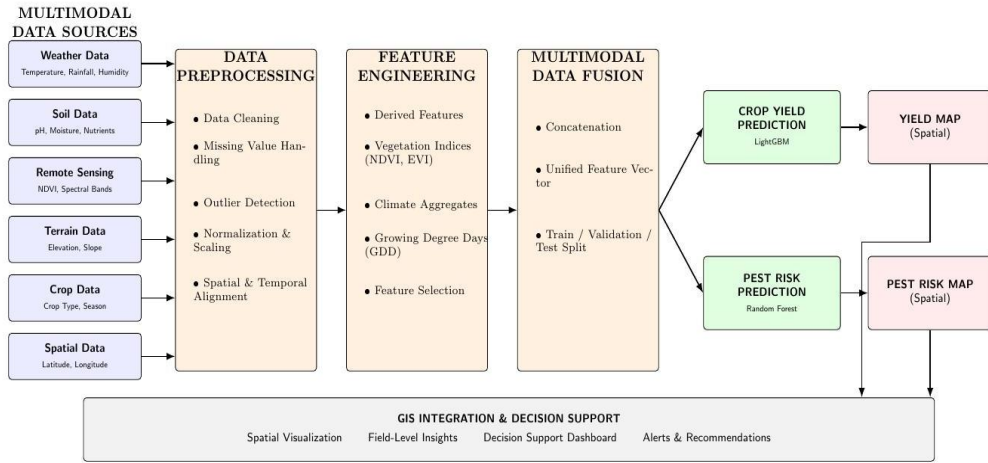


Fig. 1. Overview of the proposed AgriSense framework. Multimodal agricultural data are processed and fused for crop-yield prediction using LightGBM and pest-risk assessment using Random Forest, followed by GIS-based spatial visualization and decision support.

B. Data

AgriSense relies on six categories of heterogeneous agricultural data for crop-yield and pest-risk prediction: weather, soil, remote-sensing, terrain, crop, and spatial information. The factors like temperature, rainfall, and humidity determine the climate in which the field is situated. Soil characteristics such as its pH level, moisture content, and nutrient composition define the characteristics of the field and set the upper ceiling for the potential of the crop to perform. Remote-sensing data, including vegetation indices such as NDVI [10], reflects the current state of the canopy and serves as an in-season proxy for crop vigor, complementing the more static soil and terrain descriptors. Spatial information such as latitude, longitude, and field boundaries anchors every observation to a specific location, which is ultimately what enables AgriSense to move from a tabular prediction to a georeferenced map layer. This combination aligns with earlier multimodal agricultural research, which has demonstrated both the benefit of combining weather, soil, satellite, and terrain information for yield prediction and the fact that each modality’s importance shifts with crop and region rather than remaining fixed [3].

For crop-yield prediction, the yield observation serves as the target variable, with the remaining agricultural variables acting as predictive features. For pest-risk prediction, environmental and crop-related observations are instead used to classify the risk level. The exact datasets, geographic coverage, and observation periods will be reported once dataset selection is finalized (TBD in the camera-ready version). Table I lists the six data modalities

data — enter a shared preprocessing stage where they are cleaned and lined up with each other. From these cleaned inputs, a feature-engineering stage builds higher-level agronomic signals such as GDD, vegetation indices, and climate aggregates. All of this is concatenated into one feature vector and split into training, validation, and test sets. From there, two separate models take over: LightGBM handles the continuous yield-regression task, and Random Forest handles pest-risk classification. Both feed into a GIS layer, which turns their outputs into field-level yield and pest-risk maps and powers a decision-support dashboard with alerts and recommendations. Keeping the two modeling tasks separate lets each be tuned and updated on its own schedule, but since they share the same data-fusion backbone, preprocessing is not duplicated and the two sets of outputs stay aligned in space and time.

considered by AgriSense, their representative variables, and their role in the framework.

TABLE I
DATA MODALITIES USED IN AGRISENSE

Data Type	Representative Variables	Purpose in AgriSense
Weather	Temperature, rainfall, humidity	Yield and risk prediction
Soil	pH, moisture, nutrients	Yield and risk prediction
Remote Sensing	NDVI, spectral features	Yield prediction
Terrain	Elevation, slope	Yield and risk analysis
Crop	Crop type, season	Yield and risk prediction
Spatial	Latitude, longitude, field boundary	GIS integration
Derived	GDD, cumulative rainfall	Prediction improvement

C. Data Preprocessing

Because the collected datasets differ in format, measurement units, and spatial and temporal resolution, a preprocessing stage is required to bring them into a consistent form for modeling. Duplicate and invalid observations are dropped. Missing values are either imputed or removed, depending on how much of a variable is absent and what type it is. Numerical features are standardized where helpful; this matters mainly for the linear and

neighborhood-sensitive baselines, since tree ensembles such as LightGBM and Random Forest are largely unaffected by monotonic feature scaling. Categorical variables such as crop type and season are encoded into machine-readable form. Weather, soil, remote-sensing, and spatial observations are then aligned by location and time period, so that each training row describes the same field over the same window across every modality. The processed features are combined into a single dataset for multimodal modeling of crop yield and pest risk.

Two of these steps warrant closer examination, as they determine how much of the raw record survives into training. Missing-value handling is applied per variable rather than per row: when only a small fraction of a variable's values are absent, the gap is filled and the observation retained; when a variable is missing for most of a field-season, it is dropped rather than reconstructed, since imputing it at that density would manufacture the very signal the model is meant to learn. Rows missing the yield observation are removed outright, since that is the regression target and cannot be filled in without circularity. Outlier detection is applied to the continuous weather and soil variables, where sensor faults and transcription errors can produce readings outside any agronomically plausible range. Such values are flagged as missing rather than clipped to a boundary, so that a faulty reading is not silently converted into a plausible-looking one.

Spatial and temporal alignment is the step most specific to this setting, and it is what makes the multimodal join possible at all. The six modalities arrive on different supports: weather as station or gridded time series, remote sensing as periodic raster acquisitions, soil as sparse survey points, and crop records as per-field seasonal entries. Each is resampled onto a common unit of analysis — one field over one growing window — before the join takes place. Weather and rainfall are aggregated over the window, remote-sensing bands are averaged within the field boundary across the acquisitions that fall inside it, and soil values are drawn from the nearest survey point. Because every row is keyed on a field identifier and a growing window, that same key is later used to map a prediction back to a location in the GIS layer.

D. Feature Engineering

Beyond the raw modalities listed in Table I, AgriSense derives a small set of engineered features carrying strong agronomic signal. Growing Degree Days (GDD) come first. They accumulate the thermal energy available to a crop over a given period, with the daily value computed as

$$\text{GDD}_d = \max\left(0, \frac{T_{\max,d} + T_{\min,d}}{2} - T_{\text{base}}\right) \quad (1)$$

where $T_{\max,d}$ and $T_{\min,d}$ are the daily maximum and minimum temperatures on day d , and T_{base} is a crop-specific base temperature below which development is treated as negligible.

Summed over a growing window of D days, this yields the cumulative GDD,

$$\text{GDD}_{\text{cum}} = \sum_{d=1}^D \text{GDD}_d \quad (2)$$

which serves as a derived feature summarizing how far a crop should have progressed through its development cycle, independent of calendar date. Cumulative rainfall over the same window is computed analogously to represent water availability, and vegetation-index features such as NDVI are aggregated to the field level from the underlying remote-sensing rasters. A feature-selection step then removes redundant or low-variance predictors before the unified feature vector reaches the two models, which

keeps training time manageable and reduces the chance of the yield regressor overfitting to noisy or collinear spectral bands.

Two aspects of (1) deserve explicit mention, since they are what make GDD behave agronomically rather than as an arbitrary sum of temperatures. The outer max ensures that the contribution of each day does not go below zero; i.e., if the temperature of the day is less than the base temperature, then there is no contribution from that day. Otherwise, the total accumulation could be reduced to zero by the cold spell. The base temperature itself is crop-specific and taken from established agronomic values for the crop in question rather than fitted from the data. Fixing it in advance keeps the feature interpretable and prevents it from absorbing yield signal that properly belongs to the model. The growing window D runs from the recorded sowing date to the observation date, so that cumulative GDD at any point reflects the thermal history a particular field has actually experienced — a quantity that calendar date only approximates. The subsequent feature-selection step removes predictors that are near-constant across the sample and, among pairs of strongly correlated spectral bands, keeps a single representative; vegetation indices derived from overlapping bands are the main source of this redundancy.

E. Machine-Learning Models

Two machine-learning approaches carry out the primary prediction tasks: LightGBM for crop-yield prediction, framed as a regression problem [11], and Random Forest [12] for pest and disease-risk classification. LightGBM builds an ensemble of decision trees through gradient boosting using a leaf-wise growth strategy. Random Forest operates differently, combining many independently grown trees — each trained on a bootstrap sample using a random subset of features — so that averaging reduces variance and makes classification more robust. GDD is included as an engineered feature, as described in Section II-D, to capture temperature-dependent crop development. Both models are trained on the processed multimodal features and evaluated using task-specific metrics.

1) *Crop-Yield Prediction*: Crop-yield prediction is framed as a regression problem: the integrated agricultural features are used to estimate the expected crop yield $\hat{y}$ for a given field and season. LightGBM is chosen as the primary model here because it efficiently learns nonlinear relationships and higher-order interactions among heterogeneous features while remaining comparatively inexpensive to train on tabular data. Broadly, LightGBM constructs an additive ensemble of M regression trees,

$$\hat{y} = \sum_{m=1}^M f_m(x), \quad f_m \in \mathcal{F} \quad (3)$$

where x is the concatenated weather, soil, remote-sensing, crop, spatial, and derived feature vector for a given observation, and $\mathcal{F}$ is the space of regression trees. Trees are added sequentially to minimize a regularized objective of the form

$$\mathcal{L} = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(f_m) \quad (4)$$

where $\ell(\cdot, \cdot)$ is a differentiable loss function — squared error in this work — N is the number of training observations, and $\Omega(\cdot)$ penalizes tree complexity to control overfitting. Using the weather, soil, remote-sensing, crop, spatial, and derived features, the model produces a continuous yield estimate, which we compare against Linear Regression, Random Forest [12], and XGBoost [13] baselines using MAE, RMSE, and R^2 (Section III-A).

2) *Pest-Risk Prediction*: Pest and disease-risk prediction is framed as a classification problem over environmental and crop-related features. Random Forest is well suited to this task: it captures nonlinear relationships among many agricultural variables and remains robust to the noisy, partly redundant inputs typical of field-collected agronomic data. Using temperature, rainfall, humidity, soil conditions, and crop information, the model assigns a risk category from discrete levels (low, medium, and high), following the labels available in the dataset. A majority vote over T independently trained trees,

$$\hat{c} = \text{mode}_{t=1, \dots, T} h_t(x) \quad (5)$$

produces the final predicted class $\hat{c}$ for an input x , where $h_t(x)$ is the class chosen by the t -th tree in the forest. We measure its performance using accuracy, precision, recall, and F1-score, reported in Section III-B.

F. GIS Integration and Decision Support

Each model prediction is linked back to the spatial locations within the field that exist in the spatial database. This enables field-specific yield predictions and pest risks to be mapped out as spatial layers: a map of yield predictions and a map of pest risks (Fig. 1) rather than presented in tabular form. This is how a prediction becomes actionable for a farmer who can now identify low-yielding sections of the field or high pest risk areas and send someone to investigate them.

The dashboard automatically gives out alerts based on threshold values, such that an area or sub-area which exceeds a risk value will alert without having someone check the tables manually.

G. Evaluation Metrics

Yield prediction is scored using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), along with the coefficient of determination R^2 :

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (7)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (8)$$

where y_i and $\hat{y}_i$ are the observed and predicted yield for observation i , and $\bar{y}$ is the mean observed yield across the N evaluation samples. Smaller MAE and RMSE, together with an R^2 closer to 1, indicate closer agreement between predicted and observed yield.

Pest-risk classification is treated as a three-class problem (Low, Medium, High risk). Accuracy, precision, recall, and F1-score are computed per class and combined via macro-averaging — an unweighted mean across the class set $C = \{\text{Low, Medium, High}\}$:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{c}_i = c_i] \quad (9)$$

$$\text{Precision}_{\text{macro}} = \frac{1}{|C|} \sum_{c \in C} \frac{TP_c}{TP_c + FP_c} \quad (10)$$

$$\text{Recall}_{\text{macro}} = \frac{1}{|C|} \sum_{c \in C} \frac{TP_c}{TP_c + FN_c} \quad (11)$$

where, for a given class $c \in C$, TP_c , FP_c , and FN_c denote the true positives, false positives, and false negatives obtained by treating class c as positive and all other classes as negative (the standard one-vs-rest decomposition of the multi-class confusion matrix); c_i is the true class of observation i , and $\hat{c}_i$ is the class predicted for it by (5). Macro-averaging weights each class equally regardless of frequency, which is appropriate here: a rare High-risk field is at least as important to detect as a common Low-risk one. Precision and recall aren't just abstract numbers here — a false negative means a high-risk field goes unnoticed and the response is delayed, while a false positive sends someone out to scout or treat a field that didn't need it. F1-score is included as a single number that balances the two, reported alongside precision and recall themselves.

$$F1_{\text{macro}} = 2 \cdot \frac{\text{Precision}_{\text{macro}} \cdot \text{Recall}_{\text{macro}}}{\text{Precision}_{\text{macro}} + \text{Recall}_{\text{macro}}} \quad (12)$$

H. Training and Evaluation Protocol

As illustrated in Fig. 1, the combined feature vector is divided into training, validation, and test datasets. The models are trained on the training dataset, the validation dataset is employed to choose between the models, while the test dataset remains unused until the outcomes are known to be written up. As such, by not touching this third partition at all during the process of training, it will be possible to interpret the values presented in Section III as the estimates of the model performance on unseen data rather than its ability to fit the tuning dataset.

How the split is constructed matters more here than in a conventional tabular problem, because the rows are not independent. Two fields in the same district during the same season share weather conditions, and often soil, so a purely random split risks placing near-duplicate rows on both sides of the partition and inflating the reported score through memorization of local conditions. The partitions are therefore constructed so that observations sharing a location and growing window remain together, keeping the evaluation closer to the situation a deployed system would face: predicting for a field, or a season, the model has not previously seen. The same partitioning scheme is reused for both tasks, so that the yield regressor and the risk classifier are always evaluated on the same held-out fields, keeping their outputs comparable when placed together on a common map by the GIS layer described in Section II-F.

Both models expose hyperparameters that materially affect fit, including the number and depth of trees, the learning rate for gradient boosting, and the number of features considered at each split. These are selected on the validation partition using the task-appropriate criterion from Section II-G — RMSE for yield and macro-averaged F1-score for risk — and the chosen configuration is then refit before the single test-set evaluation. Macro-averaged F1 is used both for selection and reporting, so that the chosen classifier is the one that handles the rarer High-risk class well, rather than one that maximizes raw accuracy by favoring the majority class. The baselines discussed in Section III follow the same tuning procedure and use the same partitions, so the comparison in Table II reflects differences between model families rather than differences in tuning effort.

TABLE II
CROP-YIELD PREDICTION PERFORMANCE

Model	MAE	RMSE	R ²
Linear Regression	0.84	1.12	0.68
Random Forest	0.48	0.65	0.83
XGBoost	0.36	0.49	0.89
LightGBM (proposed)	0.27	0.38	0.93

III. RESULTS & EVALUATION

A. Crop-Yield Prediction

Crop-yield prediction is evaluated as a regression task using the multimodal agricultural features described in Section II. LightGBM is compared against three baselines — Linear Regression, Random Forest, and XGBoost — using MAE, RMSE, and R² to capture prediction error and the agreement between predicted and observed yield. Table II presents the results; lower MAE and RMSE, and higher R², indicate better performance.

LightGBM achieves the lowest MAE (0.27) and RMSE (0.38) and the highest R² (0.93) among the four models. The resulting ordering — Linear Regression < Random Forest < XGBoost < LightGBM — matches expectations: gradient-boosted tree ensembles capture nonlinear interactions among weather, soil, remote-sensing, and terrain features better than a single bagged ensemble such as Random Forest or a purely linear model. The gap between Linear Regression and the three tree-based models is substantial, with MAE dropping from 0.84 to at most 0.48, indicating that the yield signal is predominantly nonlinear. This is consistent with the multimodal, interaction-rich input space described in Section II.

B. Pest-Risk Prediction

For the purpose of predicting pest risk, four classifiers — Logistic Regression, Decision Tree, XGBoost, and Random Forest — are analyzed for their accuracy, precision, recall, and F1-Score values. These findings are depicted in the form of a grouped bar graph in Fig. 2.

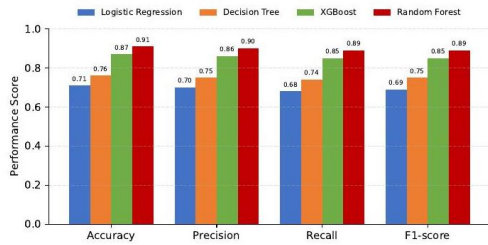


Fig. 2. Pest-risk model performance comparison across accuracy, precision, recall, and F1-score. Random Forest (proposed) achieves the best overall balance between precision and recall.

Algorithm 1 Progressive multimodal ablation protocol

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1:  $M \leftarrow \{\text{weather, soil, remote sensing, terrain, GDD}\}$ 
2:  $S \leftarrow \emptyset$ 
3: for each modality  $m \in M$  in order of expected importance
   do
4:    $S \leftarrow S \cup \{m\}$ 
5:   Build feature vector  $x_S$  using only modalities in  $S$ 
6:   Train LightGBM on  $x_S$  for yield; Random Forest on
      $x_S$  for risk
7:   Evaluate on the held-out test split using (6)–(12)
8:   Record performance for configuration  $S$ 
9: end for
10: return performance trajectory over all configurations  $S$ 

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Random Forest scores better on all four evaluation metrics — accuracy of 0.91, precision of 0.90, recall of 0.89, and F1-score of 0.89 — with the closest competitor XGBoost lagging a bit with accuracy of 0.87, precision of 0.86, recall of 0.85, and F1-score of 0.85. The only linear classifier considered, logistic regression, clearly performs worse on all four metrics (with accuracy of 0.71) — once again the same nonlinearity effect seen in the yield regression in Section III-A. For Random Forest, precision (0.90) and recall (0.89) are nearly the same values. This is critical for a pest risk advisory application as this demonstrates that Random Forest reaches the high accuracy not at the expense of one metric for another. Therefore, Random Forest is selected as the pest risk classifier in the AgriSense workflow described in Fig. 1.

C. Multimodal Analysis

To assess the contribution of each data modality, we progressively combine weather, soil, remote-sensing, terrain, and GDD features, retraining the yield and risk models on each successive combination following the ablation strategy in Algorithm 1. Comparing single-modality and combined configurations reveals whether multimodal integration actually improves crop-yield prediction and isolates the marginal contribution of each feature group; in particular, it tests whether the GDD-derived feature in (2) adds accuracy beyond what the raw temperature and rainfall variables already provide. The numerical ablation results — per-modality MAE/RMSE/R² and the corresponding classification metrics — will be reported once the final evaluation dataset is fixed. Algorithm 1 specifies the exact protocol that will generate these results, ensuring the reported values remain fully reproducible.

IV. DISCUSSION

The results in Sections III-A and III-B tell a consistent story. On both the regression (yield) and classification (pest-risk) tasks, tree-ensemble models capturing nonlinear feature interactions outperform the linear baselines. Gradient boosting, through LightGBM and XGBoost, has the advantage on the regression task, while Random Forest remains competitive throughout and, in this study, comes out on top for classification. This aligns with earlier comparative work on crop-yield modeling, which has consistently found that model family matters less than the availability and quality of the multimodal inputs behind it [1], [3].

Viewed as a system, the value of AgriSense lies less in any single metric and more in how the yield and risk sub-systems come together through the GIS layer of Section II-F. Because both predictions share the same spatial and temporal reference frame, a decision-support dashboard can pair a field predicted to under-yield with an elevated pest-risk label on the same sub-field, flagging that combination as a higher-priority alert than either signal would justify on its own. Such joint reasoning is difficult to achieve when yield and pest-risk models are built as separate, disconnected tools — precisely the gap identified for earlier single-task systems in Section I.

GDD is included as a derived feature (see (2)) because remote-sensing studies have shown temperature accumulation to be a more agronomically meaningful predictor of crop-development stage than calendar date or raw temperature readings alone [7]. The marginal contribution of GDD relative to the other weather features has not yet been separated out, which is precisely what the ablation protocol in Algorithm 1 is designed to measure. Yet, its inclusion remains conventional practice in agronomy, and it would be particularly relevant in situations where phenology is highly associated with temperature.

A. Limitations

Several limitations are worth stating plainly. The exact dataset composition, geographic coverage, and observation period are still being finalized, so the numbers in Table II and Fig. 2 should be read as the relative ordering of the models rather than as final, dataset-agnostic performance figures. The multimodal ablation study (Section III-C) has been specified but not yet run to completion, so the individual contribution of each modality — and of GDD in particular — remains a qualitative expectation rather than a measured quantity. Both LightGBM and Random Forest are tree-based ensembles; this keeps the cross-model comparison clean but also means deep-learning approaches, such as spatiotemporal or graph-based models that might better exploit the structured, multi-resolution nature of remote-sensing and terrain data, have not yet been tried. And, as with most agricultural machine-learning work, generalization across crops, seasons, and regions has not been externally validated, remaining an important target for future field-level testing.

V. CONCLUSION & OUTLOOK

AgriSense is an integrated multimodal framework for precision agriculture that brings together weather, soil, remote-sensing, terrain, crop, and spatial information for crop-yield prediction and pest-risk assessment. Yield prediction is performed with LightGBM and pest-risk classification with Random Forest, with Growing Degree Days (GDD) contributing information about crop development. On the data considered here, LightGBM outperformed the Linear Regression, Random Forest, and XGBoost baselines for yield estimation (MAE 0.27, RMSE 0.38, R^2 0.93), and among the classifiers compared, Random Forest achieved the best overall balance of precision and recall for pest-risk classification (accuracy 0.91, F1-score 0.89). By bringing these complementary data sources together with machine learning and GIS visualization, AgriSense provides one location-aware foundation for agricultural decision support.

There are a few possible avenues for further work that appear promising to explore: larger and more geographically varied datasets, integration of weather data and IoT sensors, enhanced generalizability of results for other crops and locations, and the use of explainable-AI techniques. One such technique is SHAP feature attribution, which would enable AgriSense users to see what contribution each factor (weather, soil data, remote sensing, and GDDs) made to a particular prediction. Another potential avenue is the application of AgriSense outputs to field-level validation and recommendations for irrigation, fertilizer, and pest control, as well as completing the multimodal ablation experiment from Algorithm 1.

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